02 · Environment Setup and Angular CLI

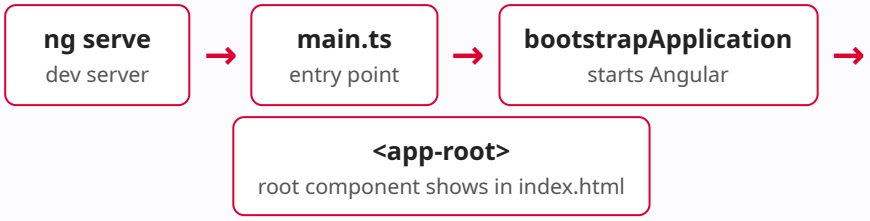
Beginner Angular learning notes

a. Definition

To build Angular apps you need **Node.js** (runs JavaScript tools on your computer), **npm** (installs packages) and the **Angular CLI** (the `ng` command). The CLI creates projects, generates components, starts a dev server and builds your app for production.

A new project contains `src/main.ts` (the entry point that starts the app), `src/app/app.ts` (the root component), `src/app/app.config.ts` (app-wide settings/providers) and `angular.json` (build settings).

b. Diagram



What happens when you run the app

c. Syntax

```
# install tools (once)
npm install -g @angular/cli

# create and run a project
ng new my-first-app
cd my-first-app
ng serve            # http://localhost:4200

# generate code
ng generate component product-card   # short: ng g c product-card
ng generate service data
ng build            # production build into dist/
```

d. Example Program

Programs/02_Setup_and_Angular_CLI/app.ts

```ts
import {
  ApplicationConfig,
  Component,
  provideBrowserGlobalErrorListeners,
} from '@angular/core';

// In a real project this goes in src/app/app.config.ts.
// main.ts then runs:  bootstrapApplication(App, appConfig)
export const appConfig: ApplicationConfig = {
  providers: [provideBrowserGlobalErrorListeners()],
```

```
};

// In a real project this is src/app/app.ts (the root component made by `ng new`).
@Component({
  selector: 'app-root',
  template: `
    <h1>Welcome to {{ appName }}</h1>
    <p>Created with: ng new {{ appName }}</p>
    <p>Started with: ng serve</p>
  `,
})
export class App {
  appName = 'my-first-app';
}
```

e. Output

Rendered in the browser (text on screen):

```
Welcome to my-first-app
Created with: ng new my-first-app
Started with: ng serve
```

f. Real-World Example (English + தமிழ்)

English

Opening a new shop: instead of arranging the building, electricity and shelves yourself, a contractor sets everything up in a day. `ng new` is that contractor.

தமிழ்

ஒரு புதிய கடை திறக்கும்போது கட்டிடம், மின்சாரம், தண்ணீர் இணைப்பு, அலமாரிகள் எல்லாம் முதலில் தயார் செய்ய வேண்டும். அந்த வேலையை ஒவ்வொன்றாகச் செய்யாமல், ஒரு ஒப்பந்தக்காரர் (contractor) ஒரே நாளில் எல்லாவற்றையும் அமைத்துக் கொடுத்தால் எவ்வளவு எளிது!

Angular CLI அந்த ஒப்பந்தக்காரர் போல. `ng new` என்று ஒரு கட்டளை கொடுத்தால் முழு project அமைப்பும் தயார்; `ng generate component` என்றால் புதிய அலமாரி தயார்; `ng serve` என்றால் கடை திறந்து வாடிக்கையாளர் பார்க்கலாம்.

g. Pros

- One command scaffolds a whole project
- Generators keep a consistent structure
- Built-in dev server, build and test commands

h. Cons

- Needs Node.js and many npm packages
- First install can be slow

i. Advanced / New Features

- Application builder uses esbuild + Vite (very fast builds)
- `ng new` creates standalone apps
- Vitest is the default test runner in v21

j. Common Mistakes

- Using an unsupported Node.js version. Check the Angular docs for the compatible Node versions.
- Running `ng serve` outside the project folder (it must be where `angular.json` is).
- Editing files in `node_modules` or `dist`; they are generated and your changes will be lost.
- Creating files by hand and forgetting to import them. `ng generate` avoids this.

k. Interview Questions

Q1. What is the Angular CLI?
A command-line tool (`ng`) that creates, builds, tests and serves Angular applications and generates code.

Q2. What does `main.ts` do?
It is the entry point; it calls `bootstrapApplication` to start the root component.

Q3. What is `angular.json`?
The workspace configuration file holding build, serve and test settings for projects.

Q4. What is the difference between `ng serve` and `ng build`?
`ng serve` runs a live-reloading dev server; `ng build` produces optimised static files for deployment.

l. Practice Exercises

1. Install Node.js and the Angular CLI, then run `ng version` and note the versions.
2. Create a project with `ng new demo`, run `ng serve`, and change the title in `app.ts` while the server is running.
3. Run `ng generate component hello` and list which files were created.